

Business Analysis

Market, unit economics, TCO, business model, go-to-market and impact · UrbanLens Platform v1.0

1 · Value proposition

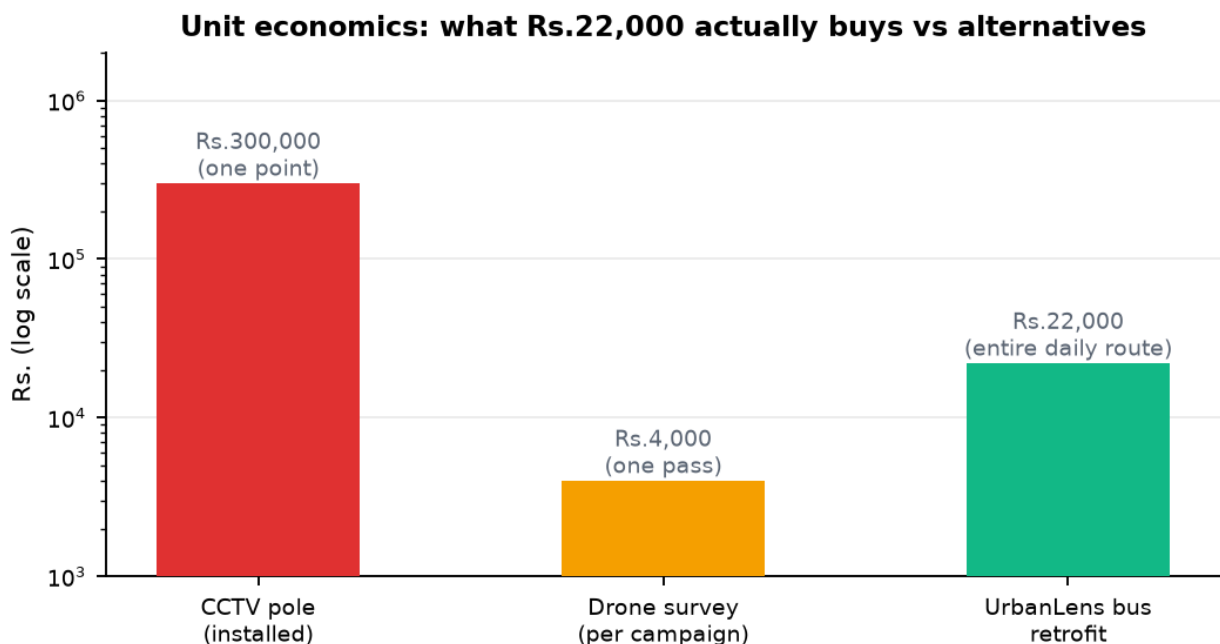
UrbanLens sells one thing: **complete, continuous road intelligence at municipal scale for near-zero marginal cost**. The customer (city) does not buy cameras, poles or surveys — it buys visibility. The sensing fleet already exists, already burns its fuel, already drives every street. We monetise the layer that turns those vehicles into infrastructure.

2 · Market sizing (India)

Layer	Estimate	Basis
TAM — smart-city road ops spend	Rs.8,000+ cr/yr	Smart Cities Mission allocations + municipal road budgets (4,000+ ULBs)
SAM — cities with 100+ bus fleets	Rs.1,200–1,800 cr/yr addressable	~100 cities/corps operating meaningful fleets; defect+traffic monitoring budgets
SOM (3-yr, realistic)	Rs.40–90 cr ARR	60–150 municipal deployments at Rs.25–50L/yr SaaS + hardware

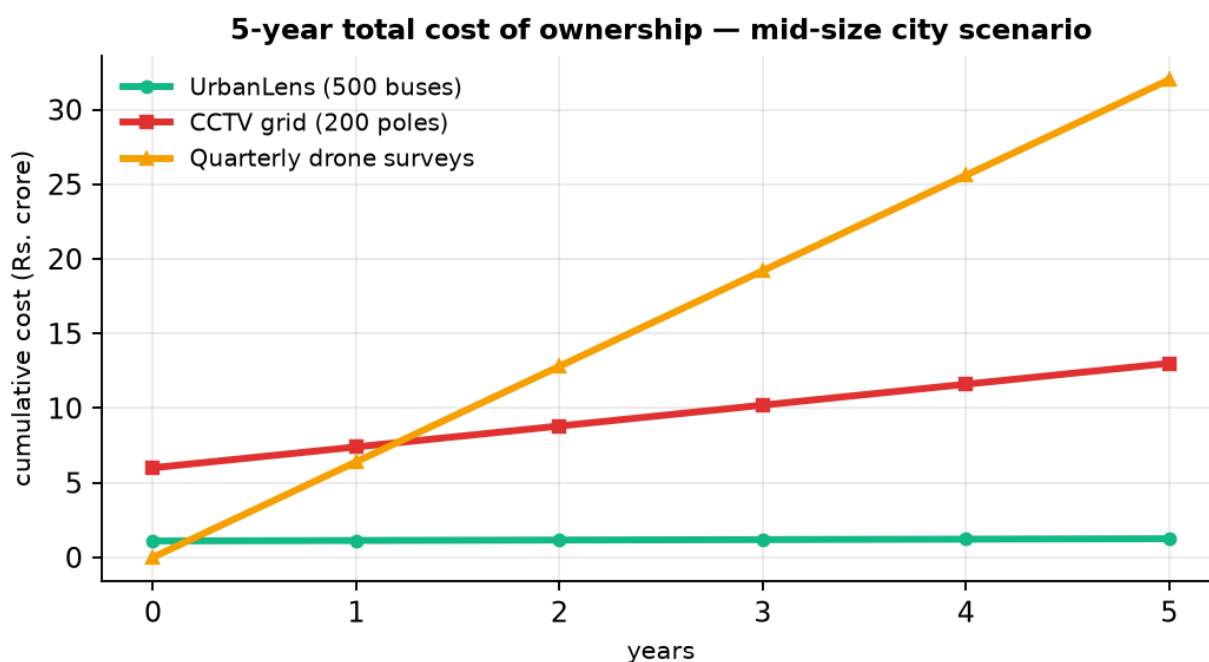
Even 1% of SAM makes this a Rs.12–18 cr/yr business — the point is the wedge, not the ceiling: once the fleet senses for roads, the same data feeds traffic planning, navigation, insurance and enforcement.

3 · Unit economics



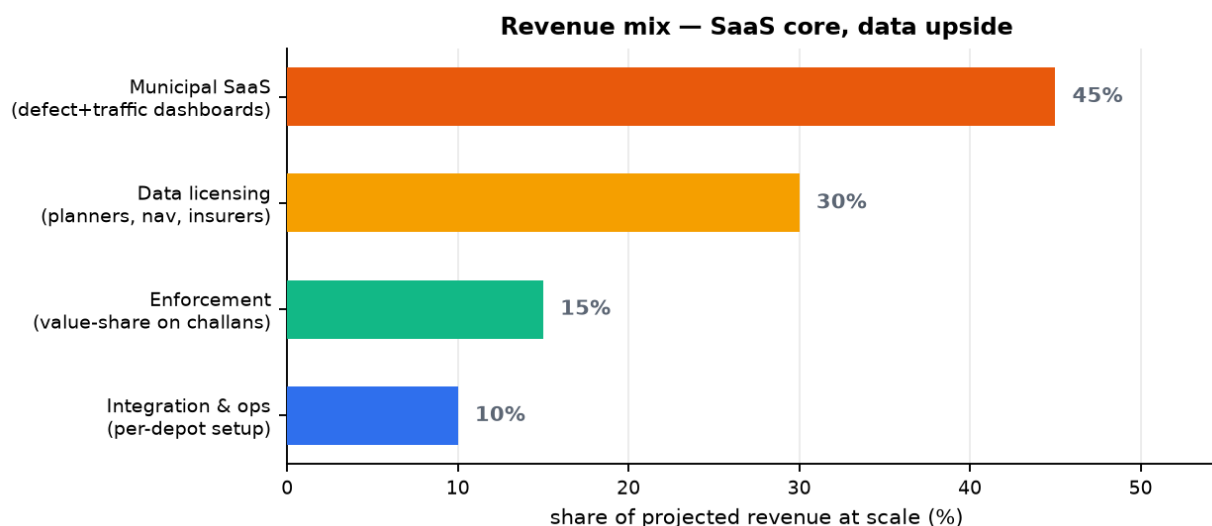
- **Hardware (one-time):** Rs.22,000/bus — Pi 5, camera, GPS, 4G dongle, housing, install.
- **Edge compute cost:** Rs.0 — all inference on-device; no cloud GPU bill ever.
- **Cloud (pilot):** Rs.0 today (free tiers); ~Rs.2,000/mo Postgres at 100-bus scale; ~Rs.25k/mo at 1,000-bus city scale.
- **Software margin:** SaaS priced per bus per month (Rs.3,000–5,000 target) against a cloud+support COGS under Rs.500 — >85% gross margin.

4 · 5-year TCO vs alternatives



A 500-bus city on UrbanLens spends ~Rs.1.25 crore over 5 years. The CCTV equivalent (200 poles) costs ~Rs.13 crore; quarterly drone surveys ~Rs.32 crore — and neither covers interior streets the buses already drive.

5 · Business model



- **Municipal SaaS (core):** per-bus subscription — dashboards, workflows, SLA tracking, support. Land via depot-level pilots (10–20 buses, 3 months, free), expand city-wide.
- **Data licensing:** anonymised traffic/OD/congestion feeds to urban planners, navigation providers, insurers (usage-based pricing), researchers.
- **Enforcement value-share:** ANPR/rash-driving intelligence for police; per-verified-case or annual licence. Positioned as evidence support, not auto-challaning.
- **Integration & ops:** depot installs, MQTT integration with existing control rooms, model updates.

6 · Go-to-market

- 1. Pilot (0–6 mo):** one depot, 10 buses, one ward engineer as champion. Success metric: defects found vs complaint-app baseline (expect 5–10× discovery rate).
- 2. City deal (6–18 mo):** city-wide rollout via transport corporation partnership — zero capex for them, revenue share on data products.
- 3. State replication (18–36 mo):** template deployments across Smart Cities Mission participants; central monitoring console for state road ministries.
- 4. Adjacent expansion:** garbage trucks (already supported fleet type), ambulances, police vehicles — the platform is vehicle-agnostic by design.

7 · Competitive landscape

Player type	Their model	Our edge
Complaint apps (Swachhata, 311)	citizens report, city reacts	proactive sensing — find before victims; no reporting fatigue
Fixed CCTV + AI vendors	junction-centric, Rs.-lakhs per pole	7–35× cheaper per sensing point; full-network coverage; no civil works
Survey consultants	periodic, manual	continuous, automated, geo-tagged evidence for every event
Fleet-telematics SaaS	GPS-only analytics	vision-based intelligence: defects, congestion causes, plates — not just movement

8 · Risks (business) & mitigations

Risk	Mitigation
Municipal sales cycles are slow	depot pilots are cheap and fast; ROI evidence (defects found/Rs. spent) closes budget conversations
Transport corporations may resist	zero capex, zero driver workload, revenue share — cooperation is financially rational
Data-privacy regulation tightens	privacy-first by architecture (on-device blur, plate-only retention); compliance is a moat, not a threat
Big tech enters the space	our cost structure (Rs.0 API, edge-only compute) is defensible at municipal price points; speed + India-specific models are the lead

9 · Impact framing (public-sector value)

- **Social:** hazards found before victims — proactive instead of reactive safety; equitable coverage of neighbourhoods that never file complaints.
- **Economic:** avoided accident and vehicle-damage costs; data-driven maintenance budgets; Rs.22k vs Rs.1–5L per sensing point.
- **Environmental:** congestion-aware routing cuts idling; garbage-route intelligence improves collection efficiency.
- **Governance:** every work order carries photo evidence, timestamp and audit trail — accountability built into the workflow.